David Yallup

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Research Themes: ***MCMC*** · ***Nested Sampling*** · ***GPU-Accelerated Inference*** · ***Generative AI for Science***

Profile

Postdoctoral researcher in probabilistic machine learning and scientific inference, specialising in GPU-accelerated Bayesian computation and generative surrogate modelling for physical simulations. Lead developer and contributor to open-source JAX scientific software, with experience translating research methods into robust, reusable code and collaborating across academia and industry.

Skills

Statistical/Generative AI: Bayesian inference; simulation-based inference; uncertainty quantification; surrogate modelling; diffusion/flow-matching models

Software: Python; JAX; PyTorch; HPC; GPU; C++; version control (Git); CI/testing; packaging; documentation

Research: Method development; benchmarking; reproducible workflows; scientific computing for cosmology/GW/HEP

Collaboration: Cross-disciplinary research; industry-facing delivery; mentoring and supervision; data science seminar organiser

Experience

Kavli Institute for Cosmology, University of Cambridge
Postdoctoral Research Associate *2021–*
Member of the Handley-lab research group:
- Developing GPU-accelerated inference for next-generation cosmology, gravitational wave, and particle physics experiments
- Building simulation-based inference and advanced MCMC tools in JAX/BlackJAX
- Bayesian statistics for scientific inference problems

Corpus Christi College, University of Cambridge
Research Associate *2022–*

Polychord Ltd.
Research Scientist *2021–*
Cambridge spin-out applying Bayesian methods to industrial problems including:
- Molecular structure prediction, battery design optimisation, Bayesian experimental design

Education

UCL
PhD Particle Physics *2015–2019*
Thesis: *"Constraining new physics with fiducial LHC measurements,"* supervised by Prof J. Butterworth.
Recipient of UCL HEP postgraduate prize.

Durham University
MSc Particles, Strings and Cosmology *2014–2015*

Durham University
MSci Natural Sciences, Maths and Physics *2009–2013*

Grants

UKRI AIRR Gateway Award: Real-time gravitational wave inference
£50k Isambard-AI GPUh credits — *2026*

Google Cloud for Researchers
$5k Google Compute Engine GPU credits — *2025*

Kavli Foundation
£3k to host Cosmological Inference in High Dimensions workshop — *2024*

Corpus Christi College, University of Cambridge
College Research Associateship — *2023*

Marie Curie Early Career Researcher
£30k for academic secondment and travel grant — *2016–2019*

Supervision

Part II Statistical Physics — **Institute of Astronomy (Maths)**
University of Cambridge — *2025*
4 groups of 2 students, ∼40 hours

Part II Relativity — **Natural Sciences Tripos (Physics)**
University of Cambridge — *2022*
4 groups of 3 students, ∼40 hours

Part III Projects — **Natural Sciences Tripos (Physics)**
University of Cambridge — *2021–*
Primary supervisor for 5 MSc projects, ∼20 hours contact each

MSci Project Supervision — **Physics and Astronomy**
UCL — *2016–2019*
Co-supervisor for 1 MSc project and 8 MSci students with Prof Butterworth

Selected Talks (2024–)

NeurIPS – FPI Workshop
San Diego — *2025*
Particle Monte Carlo methods for lattice field theory

CamGW: Cambridge Gravitational Wave Network
KICC, University of Cambridge — *2025*
Towards real-time gravitational wave inference

Future of SED Fitting
KICC, University of Cambridge — *2025*
GPU-accelerated sampling methods

ICLR – FPI Workshop
Singapore — *2025*
Nested slice sampling

Gravitational Wave Analysis in the Era of Machine Learning
Royal Astronomical Society — *2025*
GPU-accelerated sampling methods

BayesAI Workshop
Lancaster University — *2024*
Neural network advances in nested sampling

EU AI for Fundamental Physics Conference
Amsterdam — *2024*
Diffusion meets nested sampling

Cavendish Astrophysics Seminar
University of Cambridge — *2024*
Diffusion models for accelerated inference

Astrophysics ML Seminar
University of Cambridge — *2024*
Simulation-based inference

Notable publications ⋆

[1] **Yallup, David**, *Nested sampling with slice-within-gibbs: Efficient evidence calculation for hierarchical bayesian models*, 2026.

[2] N. Kroupa, **Yallup, David** and W. Handley, *Time delays and stationarity in quasar light curves*, 2026.

[3] **Yallup, David**, N. Kroupa and W. Handley, *Nested slice sampling: Vectorized nested sampling for gpu-accelerated inference*, 2026.

[4] **Yallup, David**, *Particle Monte Carlo methods for Lattice Field Theory*, in *39th Annual Conference on Neural Information Processing Systems: Includes Machine Learning and the Physical Sciences (ML4PS)*, 11, 2025 [2511.15196].

[5] D.D.Y. Ong, **Yallup, David** and W. Handley, *A Bayesian Perspective on Evidence for Evolving Dark Energy*, 2511.10631.

[6] **Yallup, David**, M. Prathaban, J. Alvey and W. Handley, *Parallel Nested Slice Sampling for Gravitational Wave Parameter Estimation*, in *2nd European AI for Fundamental Physics Conference*, 9, 2025 [2509.24949].

[7] A.N. Ormondroyd, W.J. Handley, M.P. Hobson, A.N. Lasenby and **Yallup, David**, *Dynamic or Systematic? Bayesian model selection between dark energy and supernova biases*, 2509.13220.

[8] T. Lovick, **Yallup, David**, D. Piras, A. Spurio Mancini and W. Handley, *High-Dimensional Bayesian Model Comparison in Cosmology with GPU-accelerated Nested Sampling and Neural Emulators*, 2509.13307.

[9] M. Prathaban, **Yallup, David**, J. Alvey, M. Yang, W. Templeton and W. Handley, *Gravitational-wave inference at GPU speed: A bilby-like nested sampling kernel within blackjax-ns*, 2509.04336.

[10] **Yallup, David**, N. Kroupa and W. Handley, *Nested slice sampling*, in *The Thirteenth International Conference on Learning Representations: Frontiers in Probabilistic Inference Workshop*, 2025, https://openreview.net/forum?id=ekbkMSuPo4.

[11] N. Kroupa, **Yallup, David**, W. Handley and M. Hobson, *Kernel-, mean-, and noise-marginalized Gaussian processes for exoplanet transits and H0 inference*, *Mon. Not. Roy. Astron. Soc.* **528** (2024) 1232 [2311.04153].

[12] **Yallup, David** and W. Handley, *Hunting for bumps in the margins*, *JINST* **18** (2023) P05014 [2211.10391].

[13] **Yallup, David**, W. Handley, M. Hobson, A. Lasenby and P. Lemos, *Split personalities in Bayesian Neural Networks: the case for full marginalisation*, 2205.11151.

[14] **Yallup, David**, Janßen, Timo, Schumann, Steffen and Handley, Will, *Exploring phase space with nested sampling*, *Eur. Phys. J. C* **82** (2022) 678.

[15] P. Lemos, M. Cranmer, M. Abidi, C. Hahn, M. Eickenberg, E. Massara et al., *Robust Simulation-Based Inference in Cosmology with Bayesian Neural Networks*, in *39th International Conference on Machine Learning Conference*, 7, 2022 [2207.08435].

[16] A. Buckley et al., *Testing new physics models with global comparisons to collider measurements: the Contur toolkit*, *SciPost Phys. Core* **4** (2021) 013 [2102.04377].

[17] S. Amrith, J. Butterworth, F. Deppisch, W. Liu and **Yallup, David**, *LHC Constraints on a $B-L$ Gauge Model using Contur*, *JHEP* **05** (2019) 154 [1811.11452].

[18] G. Brooijmans et al., *Les Houches 2017: Physics at TeV Colliders New Physics Working Group Report*, in *10th Les Houches Workshop on Physics at TeV Colliders*, 3, 2018 [1803.10379].

[19] J.M. Butterworth, D. Grellscheid, M. Krämer, B. Sarrazin and **Yallup, David**, *Constraining new physics with collider measurements of Standard Model signatures*, *JHEP* **03** (2017) 078 [1606.05296].

⋆ *As an ATLAS collaboration author I was an author on over 280 collaboration papers; only external small-author-list papers are listed here.*